NIQA NABILA NUR IHSANI

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SUMMARY

Computer Engineering & Informatics graduate with hands-on experience in machine learning, data analysis, and process automation. Developed a credit-risk classifier using LightGBM that achieved a 0.760 ROC-AUC and achieved 89.88% Dice and 81.85% IoU with an optimized U-Net model for MSC confluence estimation. Skilled in Python, SQL, scikit-learn, LightGBM, PyTorch, and data visualization, with experience translating technical findings into practical business and system recommendations.

TECHNICAL SKILLS

Data Science & Machine Learning: Classification (Logistic Regression, LightGBM), Deep Learning, Semantic Segmentation (U-Net), YOLO, Model Evaluation (ROC-AUC, PR-AUC, Accuracy, Precision, Recall, Dice, IoU), scikit-learn, Pandas, and NumPy.
Visualization: Tableau, Power BI, Microsoft Excel, Figma
Databases & Infrastructure: MySQL, PostgreSQL, MongoDB, Redis, Docker
Engineering & Tools: REST APIs, Agile/Scrum, Kanban, Jira, Git/GitHub, Draw.io, dbdiagram.io
Programming & Data: Python, SQL, JavaScript/TypeScript, Node.js, React.js, Next.js, Spring Boot, Django ; Talend (ETL)
Languages: Indonesian (Native); English (TOEIC 870)

WORK EXPERIENCE

Mandiri International Remittance | Remote, Kuala Lumpur
Software Engineer *Aug 2026 – Present*

- **Pipeline:** Developed an automated payment reconciliation pipeline using Django and fuzzy matching algorithms to match bank transfers with customer payments, reducing manual reconciliation.
- **Requirements:** Defined business rules, user stories, and technical requirements with Finance, Operations, and IT; evaluated implementation architecture and tracked delivery in Jira.

Meteorology, Climatology, and Geophysical Agency (BMKG) | Indonesia, Jakarta
Guest Technical Trainer *Nov 2025*

- Paid subject matter expert training 20+ senior staff on web development (HTML/CSS) and Agile (Scrum/Kanban) via live coding.

Software Engineer Intern *Jun – Oct 2025*

- **Seismic Station System (React/Node.js):** Defined 12 functional features with acceptance criteria, built 13 REST API endpoints tested in Postman, and optimized performance with Redis caching and Docker.
- **E-Learning Revitalization:** Refactored legacy PHP and redesigned database schemas (dbdiagram.io) for INATEX core modules.

Headstarter AI | Remote, New York | Software Engineering International Fellow *Jul – Sep 2024*

- Completed a 7-week fellowship building 5+ full-stack projects, including AI-powered apps, with Next.js, OpenAI, and Firebase.

PROJECT EXPERIENCE

Credit Risk Prediction & Decision Optimization – Home Credit Indonesia | Data Scientist *Feb – Mar 2026*

- **Classification Modeling:** Built ML models to predict customer default risk on a highly imbalanced dataset (91.9% non-default).
- **Model Selection:** Compared Logistic Regression and LightGBM using PR-AUC and ROC-AUC; selected LightGBM (ROC-AUC 0.760) based on offline evaluation, with an estimated 2,286 fewer incorrect rejection predictions.
- **Data-Driven Strategy:** Gain-based feature importance identified external credit scores (EXT_SOURCE_MEAN) as the strongest risk predictor; proposed tiered micro-loans and customized lending terms for higher-risk segments.

MSC Confluence Estimation via Semantic Segmentation | Machine Learning Engineer (Undergraduate Thesis) *Feb – Aug 2026*

- **Model Optimization:** Improved Dice by 1.72 and IoU by 2.76 points over baseline, reaching 89.88% Dice and 81.85% IoU.
- **Model Development:** Evaluated candidate segmentation approaches, U-Net configurations, hyperparameters, and data augmentation strategies to identify the best-performing model.
- **Generalization:** Evaluated performance across microscopy image types and applied image preprocessing to improve segmentation on non-phase-contrast images.

E-Commerce Market Intelligence System – Shopee | Data Engineer *Aug – Dec 2024*

- **System Analysis:** Defined user stories, business rules, constraints, and data requirements to plan a scalable monitoring architecture.
- **ETL & Dashboard:** Built Talend pipelines to clean and migrate scraped data into MongoDB, defined Engagement Rate logic, and co-designed analytical dashboard visualizations.

EDUCATION

Bandung State Polytechnic (Polban) | Applied Bachelor (D4) of Computer Engineering and Informatics *Aug 2022 – Aug 2026*
Relevant Coursework: Statistics and Probability, Digital Image Processing, Deep Learning, Applied Mathematics, DBMS

ORGANIZATIONAL & LEADERSHIP

Regional Youth Leadership Summit (RYLS) International 2025 | Scholarship Awardee & Delegate | Tokyo, Japan *Nov 2025*

- Awarded a competitive scholarship to represent Indonesia at RYLS 2025 in Tokyo, selected based on leadership potential and an SDG-focused essay.

Assalam Polytechnic Bandung | Deputy Head of Social Management *Mar 2023 – Jan 2025*

- Secured 20+ media partnerships for large-scale public events; managed financial planning for 7+ programs, including National PIF.

CERTIFICATIONS & AWARDS

Junior Software Quality Assurance – ArutalaLab (2026) | Data Science with Microsoft Fabric – Dicoding (2026)
Best Research Award – Literary Talent & Research Initiative (2025) | 2nd Runner-up, National SDGs Essay – Global Youth Action (2026)